EQUIPHYSICS · HORSE ARTERY LAB

The horse artery model: equations and parameters

A Python version of Vera et al. (2019), PLoS ONE *14(8): e0221425. EquiPhysics, October 2026.*

1 What the model is

The model follows blood pressure $P(x,t)$ and flow $Q(x,t)$ along 117 arteries of a resting warmblood horse, from the aortic root to the head, the four legs, and the abdominal organs (Fig. 1). A time-varying elastance left ventricle drives the tree; 62 three-element Windkessels close it. Anatomy (length, branching angle, tapered lumen diameter, and distensibility of each segment) is the study's Table 1; terminal resistances and compliances are its S1 Table. Both are open access under CC BY 4.0. The study's MATLAB code is not public; our choices where the article is silent are marked *ours* below.

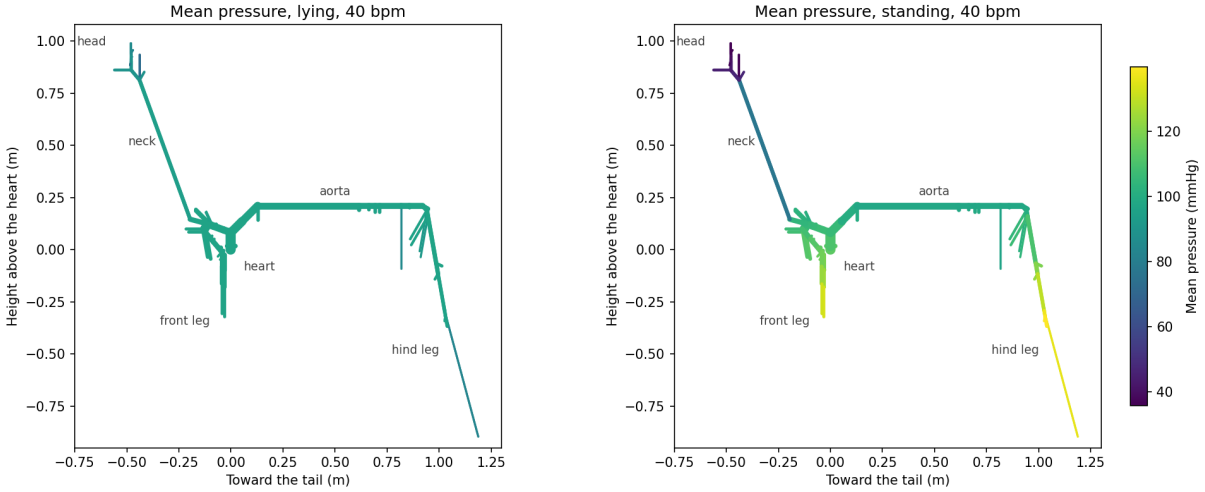


Figure 1: The 117 segments drawn from their lengths and angles (horse facing left), colored by mean pressure in a resting horse lying (left) and standing (right), on one color scale.

2 Equations in each artery

With x along the segment, A the lumen area, $R = \sqrt{A/\pi}$, and ρ, μ the blood density and viscosity,

$$C_A(x,P)\,\frac{\partial P}{\partial t} + \frac{\partial Q}{\partial x} = 0, \tag{1}$$

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\delta\,\frac{Q^2}{A}\right) + \frac{A}{\rho}\,\frac{\partial P}{\partial x} = \frac{2\pi R}{\rho}\,\tau_w - A\,g\sin\varphi. \tag{2}$$

Equation (1) is conservation of mass ($\partial A/\partial t + \partial Q/\partial x = 0$ with $A = A(x,P)$); (2) is the cross-section-averaged momentum balance. φ is the segment's angle in the side view (0° toward the tail, 90° up), so the last term is gravity along the vessel. δ is the momentum-flux coefficient: 1 (flat profile) where the Womersley number exceeds 3, 4/3 (Poiseuille) elsewhere (*ours*).

Wall law. The area compliance follows the study's Eq. (3): a reference distensibility D (Table 1, per segment) times a pressure-dependent factor,

$$C_A(x,P) = A_{\mathrm{ref}}(x)\,D\,\frac{C_p(P)}{C_p(P_{\mathrm{ref}})}, \qquad C_p(P) = a_1 + \frac{b_1}{1 + \left((P - P_{\max C})/P_{\mathrm{width}}\right)^2}. \tag{3}$$

Integrating gives the area, with $F(P) = a_1 P + b_1 P_{\text{width}} \arctan\big((P - P_{\max C})/P_{\text{width}}\big)$:

$$A(x,P) = A_{\text{ref}}(x)\left[1 + D\,\frac{F(P) - F(P_{\text{ref}})}{C_p(P_{\text{ref}})}\right]. \tag{4}$$

$A_{\text{ref}}(x)$ tapers linearly in radius between the proximal and distal diameters of Table 1. The wall stiffens as pressure rises, so the wave speed $c = \sqrt{A/(\rho C_A)}$ grows with pressure. The characteristic impedance is $Z = \rho c/A$. Below a quarter of the reference area we continue the law with a smooth exponential collapse (*ours*); normal runs never reach it.

Wall friction (Witzig–Womersley). Where $\alpha = R\sqrt{\omega\rho/\mu} > 3$ at the heart rate ω, the wall shear stress comes from the harmonics Q_n of the flow (study's Eq. 6):

$$\tau_w(t) = -\frac{4\mu}{\pi R^3}\,\bar{Q} + \sum_{n=1}^{N} \operatorname{Re}\Big\{\frac{\mu}{\pi R^3}\,\mathcal{F}(\Lambda_n)\,Q_n\,e^{in\omega t}\Big\}, \quad \mathcal{F}(\Lambda) = \frac{\Lambda\,J_1(\Lambda)/J_0(\Lambda)}{1 - 2J_1(\Lambda)/\big(\Lambda J_0(\Lambda)\big)}, \quad \Lambda_n = \alpha\sqrt{n}\,i^{3/2}. \tag{5}$$

$\mathcal{F} \to -4$ (Poiseuille) as $\alpha \to 0$ and $|\mathcal{F}| \sim \alpha$ for large α. As in the study, the harmonics come from the previous beat; we add a Poiseuille correction for the change in flow since then, which vanishes once the solution is periodic (*ours*). $N = 30$ (*ours*). Elsewhere $\tau_w = -4\mu Q/(\pi R^3)$.

3 Boundary conditions

Heart. A time-varying elastance with an internal resistance (study's Eqs. 10–11):

$$P_{\text{LV}} = E^*(t)\,\big[1 - \kappa Q(t)\big]\,(V - V_0), \qquad E^*(t) = E_{\min} + (E_{\max} - E_{\min})\,e(t), \tag{6}$$

where $e(t)$ is a normalized double-Hill curve (Stergiopulos et al. 1996, fit to Senzaki et al. 1996) peaking at $T_{\max}$ (*ours*). The aortic valve is open while $Q > 0$, and then P_{LV} equals the root pressure. When P_{LV} falls below the filling pressure the ventricle fills at $(P_{\text{fill}} - P_{\text{LV}})/R_{\text{fill}}$.

Branch points. Pressure is continuous and flow is conserved: P is the same in the parent and all daughters, and $Q_{\text{parent}} = \sum Q_{\text{daughters}}$. The equine tree has two-, three-, and four-way branchings (the carotid ends in three branches; the aorta ends in a quadrifurcation).

Terminal beds. A 3-element Windkessel (study's Eq. 7) at each of the 62 ends:

$$P = P_c + R_1 Q, \qquad C_T\,\frac{dP_c}{dt} = Q - \frac{P_c - P_{\text{out}}}{R_2}, \tag{7}$$

with total resistance $R_T = R_1 + R_2$ and compliance C_T from the S1 Table and $R_1 = Z_c$ (no reflection at high frequency). In nine small head branches $Z_c > R_T$; there $R_1 = R_T$, $R_2 = 0$ (*ours*). $P_{\text{out}} = 0$.

4 Numerical method (ours)

The study used an implicit finite-difference scheme. We use an explicit MacCormack scheme in the interior (sweep direction alternating each step) and the method of characteristics at every segment end: along $dx/dt = u \pm c$ the quantities $W_\pm = P \pm ZQ$ (plus the source term) carry information to the boundary, where they are combined with the heart, branch, or Windkessel conditions. A Jameson-type shock-capturing damping keeps the steepening pulse front from ringing. Grid spacing 10 mm (1,111 points), time step about 0.16 ms (CFL 0.8). Runs start from 100 mmHg and 1 ml/s (as in the study) and repeat beat to beat after about seven beats; we report the last beat of eight (the notebook) or ten (the comparison script), which agree to the digits shown. Halving the grid spacing changes the reported numbers by 1–2%.

5 Parameters

Group	Parameter	Value	Source
Blood	density ρ; viscosity μ	1050 kg/m^3; 0.004 Pa s	study
Gravity	g; segment angles φ	9.81 m/s^2; Table 1	study
Wall	a_1, b_1, $P_{\max C}$, P_{width}, P_{ref}	0.76, 5, 10 mmHg, 21 mmHg, 100 mmHg	study
	distensibility D; diameters; lengths	per segment, Table 1	study
Heart	heart rate; $T_{\max}$	40 beats/min; 478 ms	study (systolic duration)
	$E_{\max}$; $E_{\min}$	0.26; 0.0127 mmHg/ml	study; calibrated (study: 0.01)
	κ; V_0	55×10^{-6} s/ml; 0 ml	study
	filling pressure; filling resistance	16 mmHg; 0.003 mmHg s/ml	study
	elastance curve shape	double Hill: 0.303, 0.508, 1.32, 21.9	ours (Stergiopulos 1996)
Terminals	R_T, C_T	per terminal, S1 Table	study
	R_1; P_{out}	Z_c (or R_T if smaller); 0	study; ours
Friction	Womersley where $\alpha > 3$; harmonics	–; 30	study; ours
Numerics	Δx; CFL; damping κ_2; beats	10 mm; 0.8; 1; 8–10	ours

With every parameter as printed, the model pumps 39.6 L/min instead of the study's 33. The study's stroke volume (820 ml) and ejection fraction (65%) imply an end-diastolic volume near 1,260 ml, while $E_{\min} = 0.01$ lets ours fill to 1,530 ml. Matching that one number ($E_{\min} = 0.0127$) is our only calibration; gravity is never tuned.

6 How it compares with the study

	Study, lying	Ours, lying	Study, standing	Ours, standing
Cardiac output (L/min)	33	32.5	30	29.6
Stroke volume (ml)	820	812	740	740
Aortic root pressure (mmHg)	114/70	114/72	131/88	131/90
Pulse pressure (mmHg)	44	42	43	41
Total vascular resistance (mmHg s/ml)	0.17	0.18	0.22	0.23
Aortic pulse wave speed (m/s)	5.3	5.1	5.2	5.1

Mean flows (Reynolds numbers) at the five sites agree within 10%, Womersley numbers within 6%, and the waveform shapes and the shift of blood from head to legs on standing match the study's figures. Peak velocities differ by up to 30% at some sites, limb pulse pressures are 10–20% lower than the study's, and peak wall shear depends on how many harmonics are kept (the study does not say). Like the study's model, ours overestimates pulse pressure by about 70%: 56 vs 33 mmHg in the carotid of standing horses, and 41 vs 24 mmHg at the aortic root (measured in one anesthetized horse). Full tables: `results/comparison.md`.

7 Teaching controls

The notebook exposes three settings: *standing* (gravity on or off), *heart rate* (35–80 beats/min, with $T_{\max}$ scaled by $\sqrt{40/\text{HR}}$, ours), and *stiffness* s (every distensibility divided by s, 0.7–1.5). The model has no reflexes or autoregulation, so these show mechanics only. In a standing horse with a slow heart and stiff arteries, head pressure can fall near zero before the next beat; the head arteries soften until blood outruns the pressure wave and the 1D equations fail. The notebook reports this in plain words.

References

Vera L, Campos Arias D, Muylle S, Stergiopulos N, Segers P, van Loon G (2019). A 1D computer model of the arterial circulation in horses: an important resource for studying global interactions between heart and vessels under normal and pathological conditions. *PLoS ONE* 14(8): e0221425.

Reymond P, Merenda F, Perren F, Rüfenacht D, Stergiopulos N (2009). Validation of a one-dimensional model of the systemic arterial tree. *Am J Physiol Heart Circ Physiol* 297: H208–H222.

Stergiopulos N, Meister JJ, Westerhof N (1996). Determinants of stroke volume and systolic and diastolic aortic pressure. *Am J Physiol* 270: H2050–H2059.

Senzaki H, Chen CH, Kass DA (1996). Single-beat estimation of end-systolic pressure-volume relation in humans. *Circulation* 94: 2497–2506.

Womersley JR (1955). Method for the calculation of velocity, rate of flow and viscous drag in arteries when the pressure gradient is known. *J Physiol* 127: 553–563.

MacCormack RW (1969). The effect of viscosity in hypervelocity impact cratering. AIAA Paper 69-354.